

NESTED QUERIES

Gianluca Della Vedova

University of Milano - Bicocca

AUTHORS TABLE

id	last_name	first_name	DoB	Income	Genre
1	Lopez Baranda	Christina	15/11/2000	55000	Fantasy
2	Jin-Soon	Sin	29/03/1983	65000	Crime
3	Jones	Hannah	01/02/1973	129000	Fantasy
4	Novak	Stanislaw	12/12/1992	91000	Crime
5	Turay	Tandice	09/07/1980	99000	Romance
6	Roy	Shanta	11/10/1977	55000	Fantasy
7	Berger	Henry	15/08/1956	63000	Romance
8	Khatami	Paree	11/10/1966	86000	Sci-Fi

INTERSECT

```
SELECT last_name, first_name  
FROM Authors  
WHERE Income > 90000  
  
INTERSECT  
  
SELECT last_name, first_name  
FROM Authors  
WHERE last_name LIKE '%o%'
```

REMOVE INTERSECT WITH IN

```
SELECT last_name, first_name  
FROM Authors  
WHERE Income > 90000 AND last_name IN  
    SELECT last_name  
    FROM Authors  
    WHERE last_name LIKE '%o%'
```

REMOVE INTERSECT WITH EXISTS

```
SELECT A1.last_name, A1.first_name
FROM Authors A1
WHERE A1.Income > 90000 AND EXISTS
    SELECT *
    FROM Authors A2
    WHERE A2.id=A1.id AND A2.last_name LIKE '%o%'
```

RESULT

id	last_name	first_name	DoB	Income	Genre
3	Jones	Hannah	01/02/1973	129000	Fantasy
4	Novak	Stanislaw	12/12/1992	91000	Crime

EXCEPT

```
SELECT last_name, first_name  
FROM Authors  
WHERE Income > 90000  
EXCEPT  
SELECT last_name, first_name  
FROM Authors  
WHERE last_name LIKE '%o%'
```

REMOVE EXCEPT WITH NOT IN

```
SELECT last_name, first_name  
FROM Authors  
WHERE Income > 90000 AND id NOT IN  
    SELECT id  
    FROM Authors  
    WHERE last_name LIKE '%o%'
```

REMOVE EXCEPT WITH NOT EXISTS

```
SELECT A1.last_name, A1.first_name  
FROM Authors A1  
WHERE A1.Income > 90000 AND NOT EXISTS  
    SELECT *  
    FROM Authors A2  
    WHERE A2.id=A1.id AND A2.last_name LIKE '%o%'
```

RESULT

id	last_name	first_name	DoB	Income	Genre
5	Turay	Tandice	09/07/1980	99000	Romance

ROW ACHIEVING A MAXIMUM WITH EXISTS

Find the author(s) with maximum income

```
SELECT last_name, first_name
FROM Authors A1
WHERE NOT EXISTS SELECT *
                  FROM Authors A2
                  WHERE A1.income < A2.income
```

ROW ACHIEVING A MAXIMUM WITH IN

Find the author(s) with maximum income

```
SELECT last_name, first_name  
FROM Authors A1  
WHERE A1.income IN (SELECT max(A2.income)  
                    FROM Authors A2)
```

Except where noted, text and images for Introduction to Relational Databases by Mark Jordan is licensed under a Creative Commons Attribution 4.0 International License. Except where noted, text and images for Introduction to Relational Databases by Gianluca Della Vedova is licensed under a Creative Commons Attribution 4.0 International License. Everything in the 'scripts' directory is in the public domain (CC0).